

Informed Search: Admissible Heuristics and Domain Relaxation

Part-6

Introduction

In the realm of Artificial Intelligence, solving complex problems often hinges on the efficiency of search algorithms. While uninformed search methods systematically explore a search space, informed search algorithms leverage additional knowledge, known as a heuristic function, to guide their exploration more intelligently. This lecture delves into the critical aspect of how such heuristic functions are designed, particularly focusing on "admissible heuristics" which guarantee optimal solutions in algorithms like A*.

The challenge lies not just in having a heuristic, but in constructing one that is both effective and admissible. An admissible heuristic provides an optimistic estimate of the cost to reach the goal, ensuring that the algorithm never overestimates the true cost. This property is crucial for maintaining optimality guarantees. We will explore a powerful theoretical framework, "domain relaxation," which offers a systematic approach to deriving admissible heuristics for new problems, transforming the art of heuristic design into a more structured science.

Core Concepts

Term	Definition	Significance
Admissible Heuristic	An optimistic estimate of the cost from a given node to the nearest goal state, always providing a lower bound for minimization problems and an upper bound for maximization problems.	Guarantees optimality for algorithms like A* in tree search, ensuring the found solution is the best possible.
Optimistic Heuristic	A heuristic that estimates the goal is closer (or cheaper) than it actually is, never overestimating the true cost to the goal.	Essential for admissibility, as it prevents the search from prematurely discarding potentially optimal paths.

Term	Definition	Significance
Relaxed Problem	A simplified version of an original problem where one or more restrictions or rules have been removed.	The optimal solution cost of a relaxed problem provides an admissible heuristic for the original problem.
h1 (Misplaced Tiles)	A heuristic for the 8-puzzle problem that counts the number of tiles not in their correct goal positions.	An admissible heuristic derived from relaxing the rule that tiles can only swap with neighbors.
h2 (Manhattan Distance)	A heuristic for the 8-puzzle problem that calculates the sum of the Manhattan distances for each tile from its current position to its goal position.	A more informed and often more effective admissible heuristic than h1 , derived from relaxing the rule that tiles can only move into an empty adjacent square.
Dominance	A property where one admissible heuristic h2 is said to dominate another h1 if $h2(n) \geq h1(n)$ for all nodes n .	A dominating heuristic is generally more informative and leads to faster search times for algorithms like A*.
Uniform Cost Search	A search algorithm that expands nodes based on their cumulative path cost from the start node, effectively A* with a heuristic of $h(n) = 0$.	Represents the most uninformed admissible heuristic, demonstrating the foundational connection between different search algorithms.
Hamiltonian Cycle Problem	The problem of finding a cycle in a graph that visits each node exactly once and returns to the starting node.	Used as an example to illustrate the application of domain relaxation for deriving admissible heuristics in graph problems.
Minimum Spanning Tree (MST)	A subset of the edges of a connected, edge-weighted undirected graph that connects	Can be used to derive an admissible heuristic for problems like the

Term	Definition	Significance
	all the vertices together, without any cycles and with the minimum possible total edge weight.	Hamiltonian Cycle by relaxing constraints on node degrees and exact visitation.

Detailed Analysis

1. Revisiting Informed Search and Admissibility

Informed search algorithms, such as A*, leverage heuristic functions to estimate the cost from any given node to the goal. This additional information allows them to explore the search space more efficiently than uninformed methods. A crucial property for a heuristic, especially for A* in tree search, is admissibility. An admissible heuristic is one that never overestimates the true cost to reach the goal. For minimization problems, this means the heuristic value $h(n)$ must always be less than or equal to the actual cost $h^*(n)$. This "optimistic" nature ensures that A* can guarantee finding an optimal solution.

The lecture highlights that if an admissible heuristic is not used, the optimality guarantees of A* are lost. While suboptimal algorithms exist (e.g., IDA or depth-first branch and bound for memory efficiency), the focus here is on how to systematically derive heuristics that maintain optimality. The most uninformed admissible heuristic is $h(n) = 0$, which effectively turns A* into Uniform Cost Search, demonstrating the spectrum of heuristic informativeness.

2. Heuristics for the 8-Puzzle Problem

The 8-puzzle serves as a classic example to illustrate different admissible heuristics and their impact on search performance. The goal is to rearrange 8 numbered tiles on a 3x3 grid to a target configuration, with one empty space allowing tile movements.

- **h_1 : Number of Misplaced Tiles**
 - This heuristic counts how many tiles are not in their correct final position. For example, if 8 tiles are misplaced, h_1 would be 8.
 - **Admissibility:** h_1 is admissible because each misplaced tile requires at least one move to reach its correct position. Since only one tile can move at a time, the number of misplaced tiles provides a lower bound on the number of moves needed.
- **h_2 : Manhattan Distance**

- This heuristic calculates the sum of the Manhattan (or city-block) distances for each tile from its current position to its goal position. The Manhattan distance is the sum of the absolute differences of their x and y coordinates. For example, if a tile needs to move 1 step right and 2 steps down, its Manhattan distance is 3.
- **Admissibility:** h_2 is admissible because each tile must move at least its Manhattan distance to reach its goal. The presence of other tiles or the empty space might force more moves, but never fewer.
- **Comparison and Dominance:**
 - h_2 is generally a "better" heuristic than h_1 because $h_2(n) \geq h_1(n)$ for all nodes n . This property is called dominance.
 - A dominating heuristic is more "informed" and typically leads to significantly fewer node expansions and faster search times for A^* .
- **Performance Example (from lecture):**
 - Goal Depth 12: Iterative Deepening Search (IDS) expands 3 million nodes. A with h_1 expands 227 nodes. A with h_2 expands 73 nodes.
 - Goal Depth 24: IDS fails to find a solution. A with h_1 expands 39,000 nodes. A with h_2 expands 1,600 nodes.
 - This clearly demonstrates that a more informed (dominating) admissible heuristic drastically improves performance.

3. The General Principle: Domain Relaxation

The core idea for systematically generating admissible heuristics is domain relaxation.

- **Definition:** A relaxed problem is one derived from the original problem by removing one or more of its restrictions or rules.
- **Mechanism:** When you remove constraints, the problem becomes easier to solve. The optimal solution to this relaxed problem will always have a cost that is less than or equal to the optimal solution of the original problem. This is because any valid solution to the original problem is also a valid (though not necessarily optimal) solution to the relaxed problem. Since the relaxed problem has fewer constraints, it might have even cheaper solutions that are not valid in the original problem.
- **Admissibility:** Therefore, the cost of an optimal solution in the relaxed problem serves as an admissible heuristic for the original problem. It provides a guaranteed lower bound on the true cost.

Let's apply this to the 8-puzzle heuristics:

- **h1 (Misplaced Tiles) as a Relaxed Problem:**

- Original Rule: A tile can only swap with an adjacent empty space.
- Relaxed Rule for **h1** : A tile can move to its correct position anywhere on the board, as long as it's swapping with the empty space. (The lecture describes this as "allowing tiles to fly" to the gap).
- In this relaxed problem, each misplaced tile can move directly to its goal position in one step (by swapping with the empty space, which is assumed to be available or movable without cost). The optimal cost for this relaxed problem is simply the count of misplaced tiles.

- **h2 (Manhattan Distance) as a Relaxed Problem:**

- Original Rule: A tile can only move to an adjacent empty square.
- Relaxed Rule for **h2** : A tile can move to any adjacent square, regardless of whether it's empty or occupied. (The lecture describes this as "a tile can move to any adjacent square in spite of whether there is a tile there or not").
- In this relaxed problem, tiles can "pass through" each other. The optimal cost for moving a single tile to its goal is its Manhattan distance. Summing these for all tiles gives **h2** .

4. Domain Relaxation for the Hamiltonian Cycle Problem

To further illustrate the power of domain relaxation, consider the Hamiltonian Cycle Problem: find a cycle that visits every node in a graph exactly once and returns to the starting node.

- **Problem Definition:** A state can be represented as a partial path (e.g., **a1** -> **a2** -> **a3** -> ... -> **ak**). The goal is to complete this path into a cycle visiting all nodes.

- **Solution Constraints:**

- Visit all nodes.
- Visit each node exactly once.
- Each node in the final cycle must have a degree of exactly two (one incoming, one outgoing edge).

- **Deriving Heuristics via Relaxation:**

- **Relaxing "Visit all nodes":** One could consider the shortest path from the current end node (**ak**) back to the start node (**a1**) using only the remaining unvisited nodes. This would be an admissible heuristic, as the

actual cycle must include this path, but it might be longer due to other constraints.

- **Relaxing "Each node has degree two" and "Visit each node exactly once"**: If we relax these constraints, but still require visiting all remaining nodes, we can consider the Minimum Spanning Tree (MST) on the set of unvisited nodes (plus the current end node and start node, if applicable).
- The cost of an MST connecting all remaining nodes provides a lower bound for the cost of completing the Hamiltonian cycle. This is because any Hamiltonian path/cycle connecting these nodes must at least connect them, and an MST is the cheapest way to connect a set of nodes. The MST does not enforce degree-two constraints or cycle formation, making it a relaxed problem.
- The MST cost will always be less than or equal to the cost of the actual path segment needed to complete the Hamiltonian cycle, thus making it an admissible heuristic.

Key Takeaways

- [] Informed search algorithms use heuristic functions to guide their search, improving efficiency.
- [] An admissible heuristic never overestimates the true cost to the goal, ensuring optimality for A* in tree search.
- [] For minimization problems, an admissible heuristic provides a lower bound on the cost.
- [] The concept of "domain relaxation" is a systematic method for constructing admissible heuristics.
- [] A relaxed problem is one where some constraints of the original problem are removed, making it easier to solve.
- [] The optimal solution cost of a relaxed problem is always an admissible heuristic for the original problem.
- [] Heuristics like "misplaced tiles" and "Manhattan distance" for the 8-puzzle can be understood as solutions to relaxed versions of the problem.
- [] A dominating heuristic (one that is always greater than or equal to another admissible heuristic) is generally more informative and leads to faster search performance.
- [] Complex problems like the Hamiltonian Cycle can leverage relaxed problems (e.g., Minimum Spanning Tree) to derive admissible heuristics.

Conclusion

The ability to design effective and admissible heuristic functions is paramount for the practical application of informed search algorithms. This lecture has illuminated that heuristic design is not merely an intuitive art but can be approached systematically through the principle of domain relaxation. By identifying and strategically removing constraints from a problem, we can create simpler, "relaxed" versions whose optimal solutions provide guaranteed lower bounds—and thus admissible heuristics—for the original, more complex problem.

This framework allows AI practitioners to move beyond ad-hoc heuristic creation, offering a powerful tool for developing intelligent agents that can solve challenging problems efficiently and optimally. Understanding domain relaxation not only helps in crafting better heuristics but also deepens our comprehension of the underlying structure and constraints of the problems we aim to solve, paving the way for more sophisticated and robust AI solutions.